In [5]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

In [10]:
```
heart = pd.read_csv('Heart diseases prediction.csv')
```

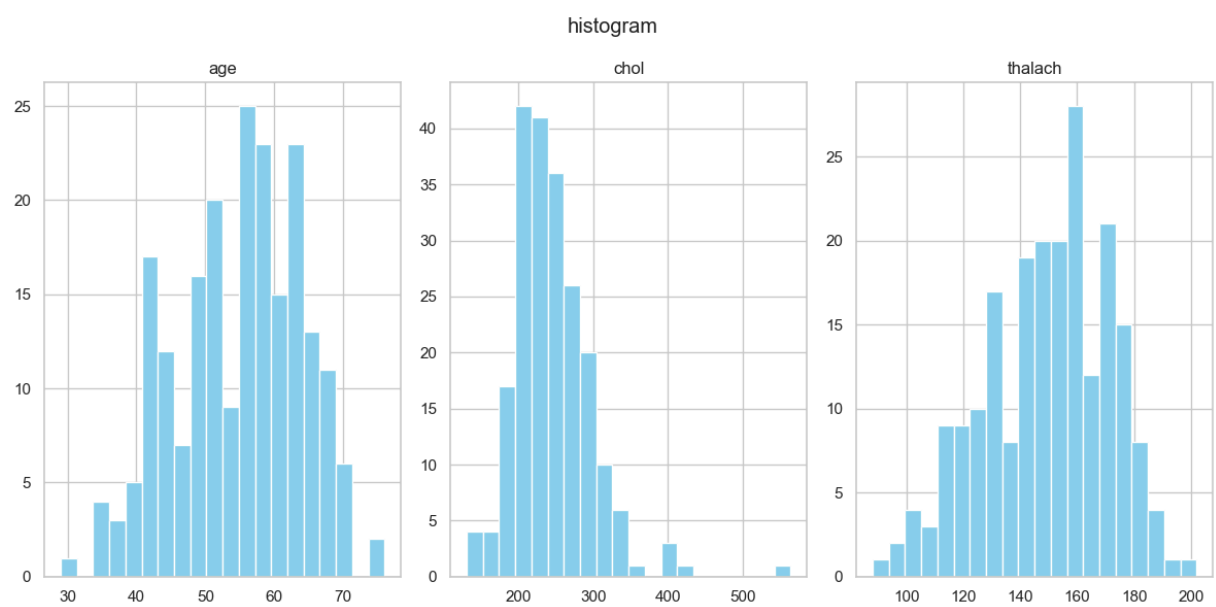
In [12]:
```
print("columns in the dataset:")
print(heart.columns)
```

```
columns in the dataset:
Index(['age', 'sexe', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach',
       'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],
      dtype='str')
```

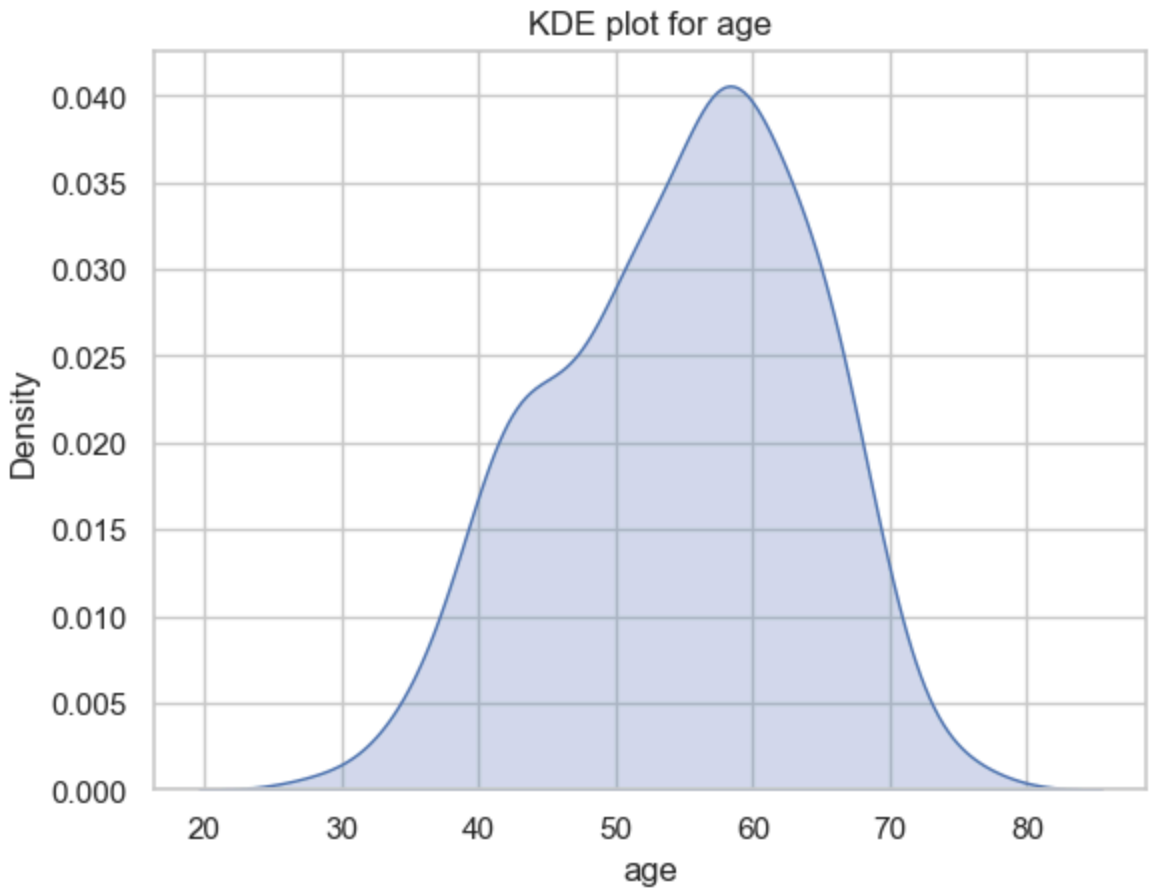
In [13]:
```
heart['binary_target'] = heart['target'].apply(lambda x: 0 if x ==0 else 1)
```

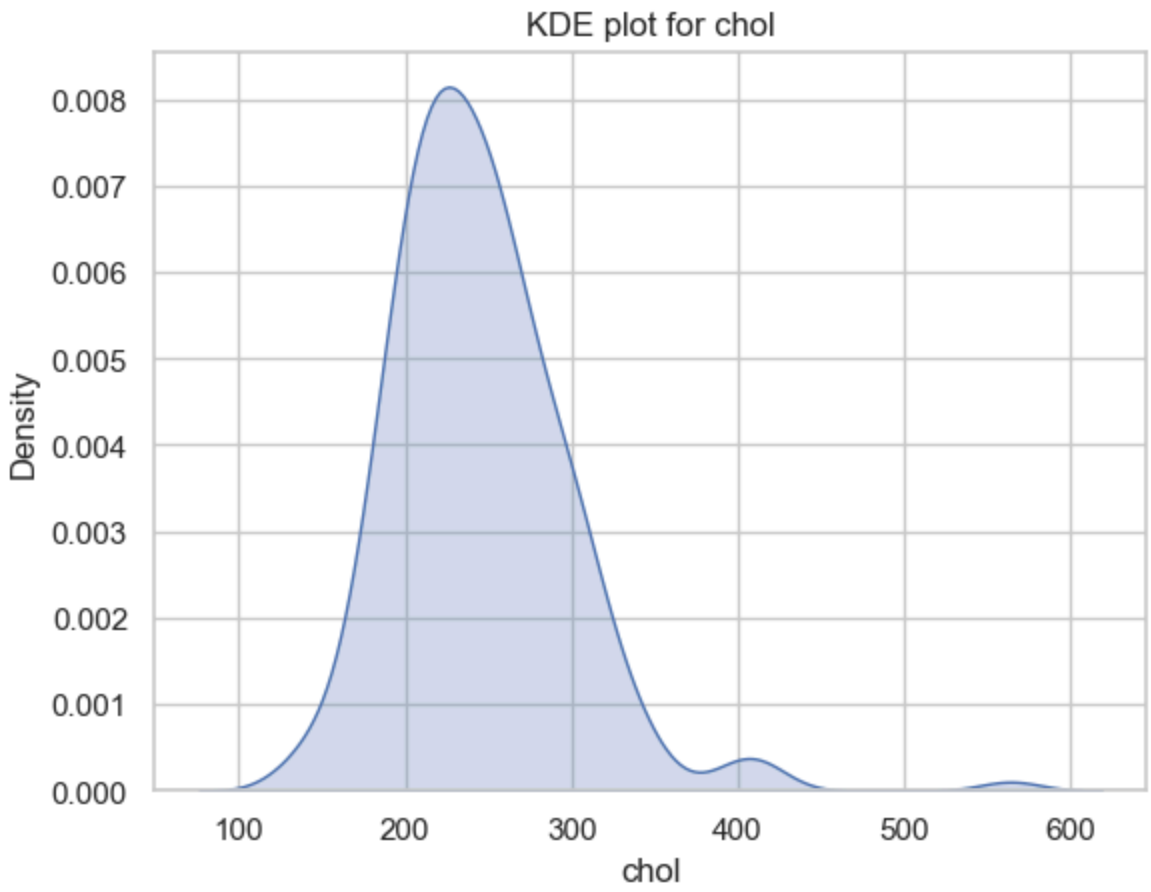
In [14]:
```
sns.set(style="whitegrid")
```

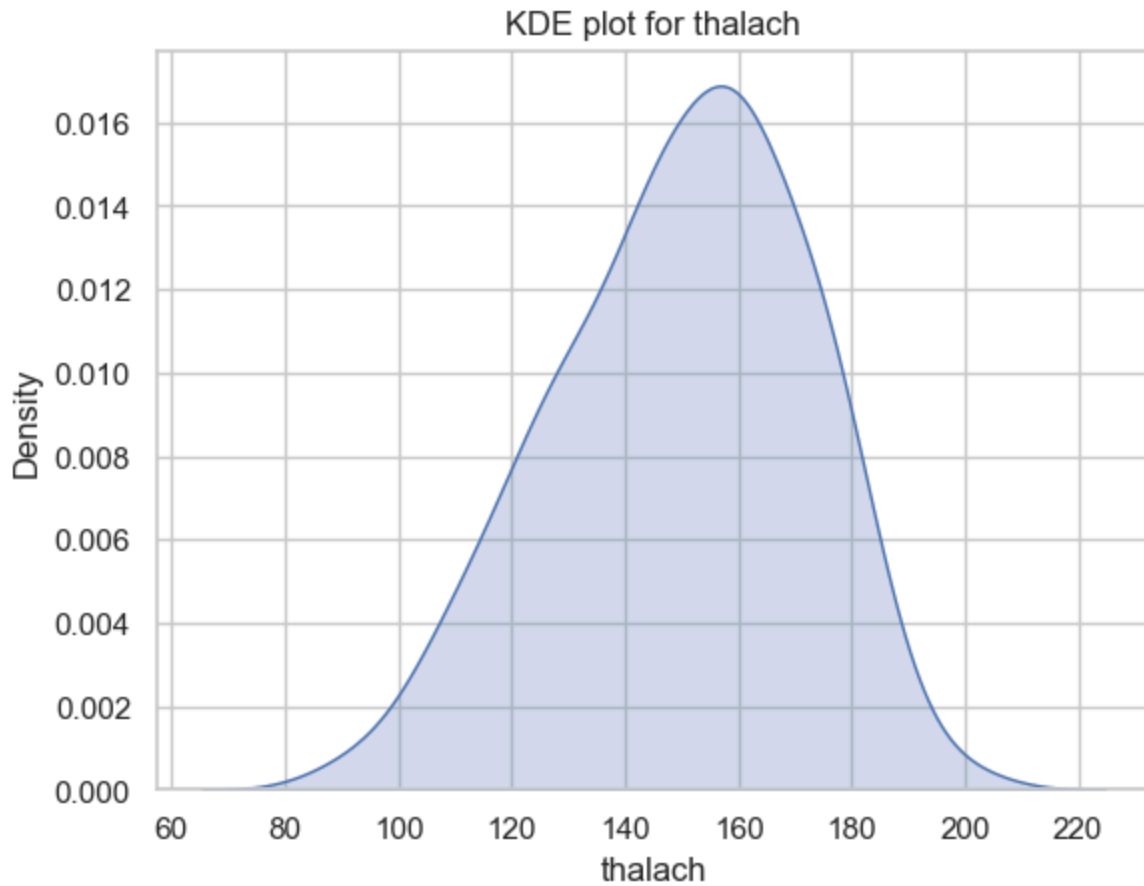
In [23]:
```
#Histogram
heart[["age","chol","thalach"]].hist(bins=20, figsize=(12,6), layout=
(1,3), color='skyblue')
plt.suptitle("histogram")
plt.tight_layout()
plt.show()
```



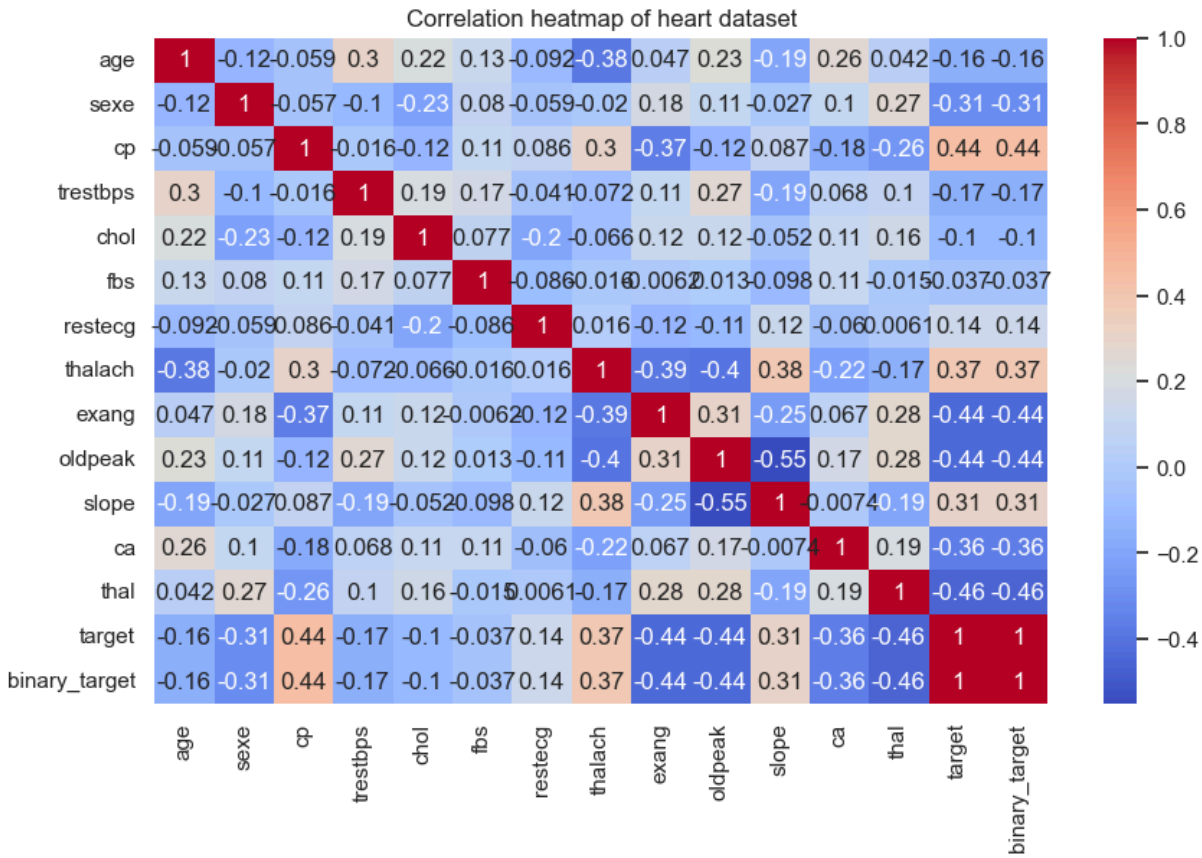
In [24]:
```
#KDE plot
for column in ["age", "chol", "thalach"]:
 sns.kdeplot(data=heart[column], fill=True)
 plt.title(f"KDE plot for {column}")
 plt.xlabel(column)
 plt.ylabel("Density")
 plt.show()
```


KDE plot for age
Density
0.040
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
20
30
40
50
60
70
80
age

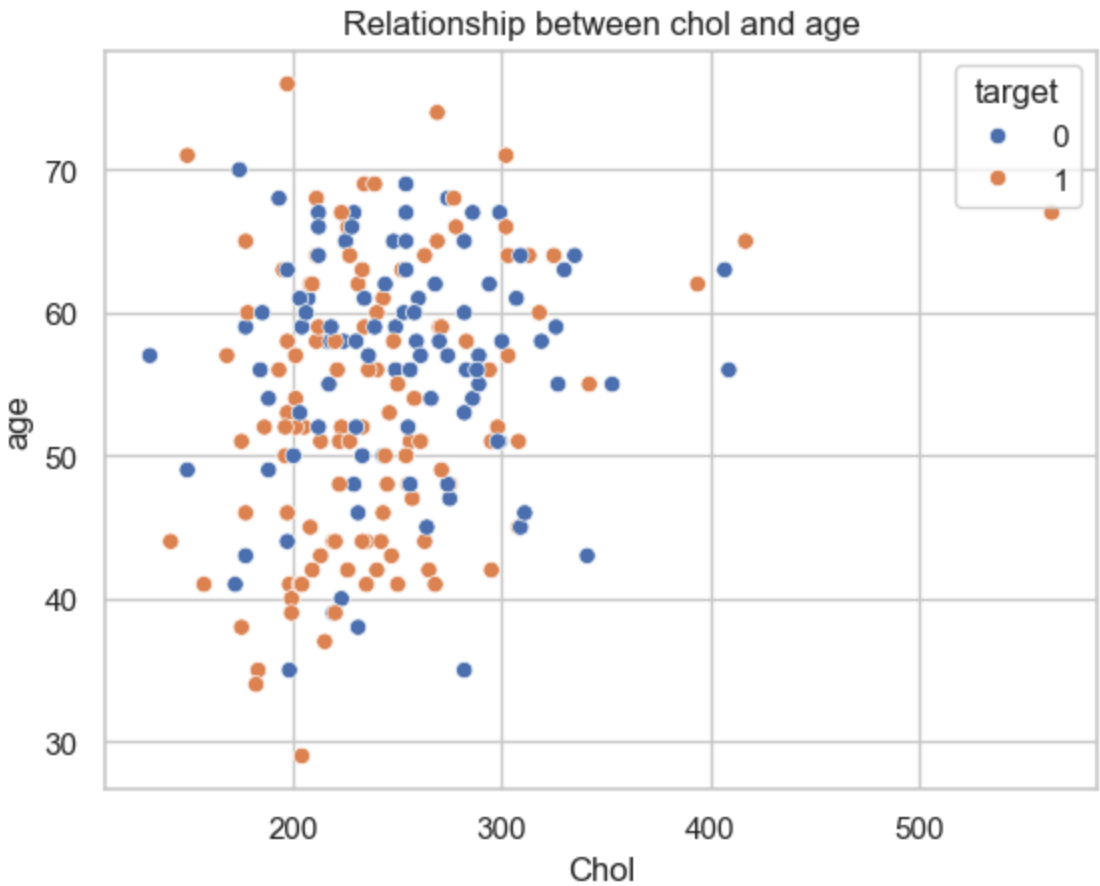

KDE plot for chol
Density
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0.000
100
200
300
400
500
600
chol



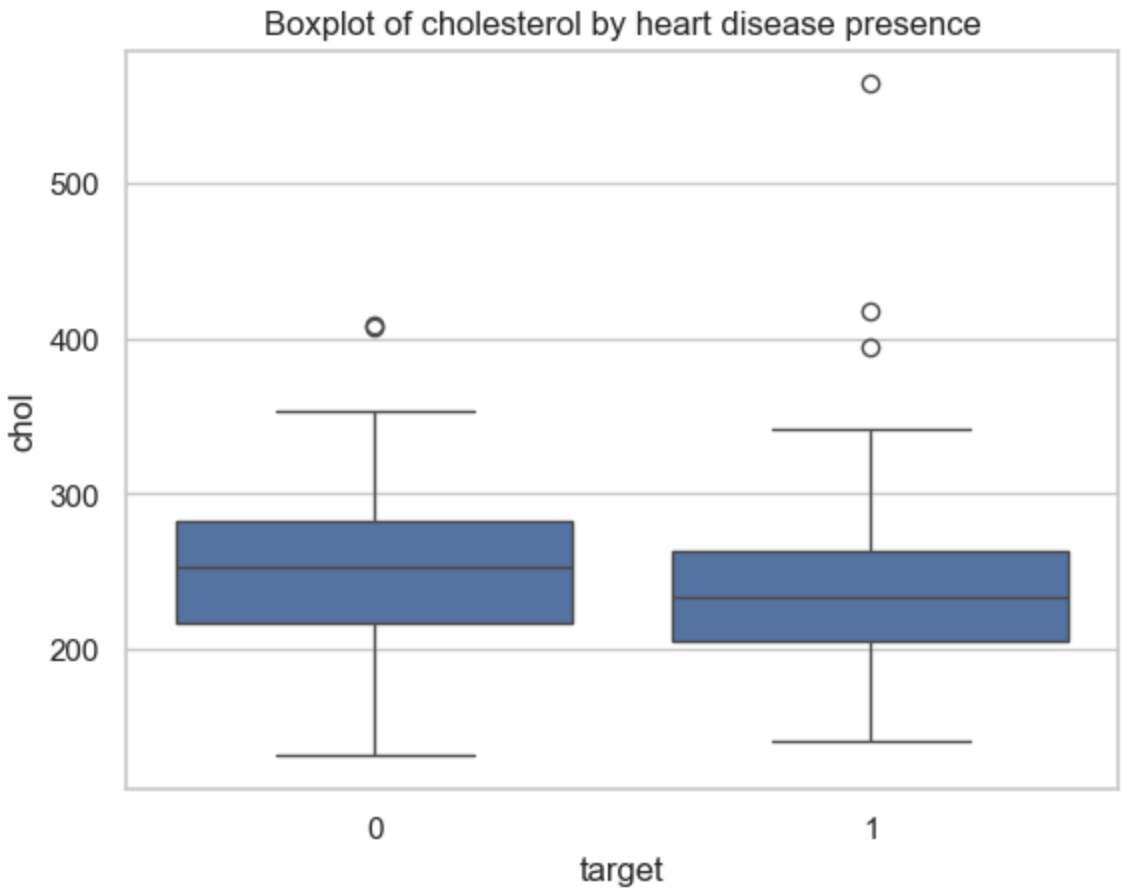
In [25]:
```
#Heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(heart.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation heatmap of heart dataset")
plt.show()
```



In [33]:
```
sns.scatterplot(x='chol', y='age', hue="target", data=heart)
plt.xlabel("Chol")
plt.ylabel("age")
plt.title("Relationship between chol and age")
plt.show()
```



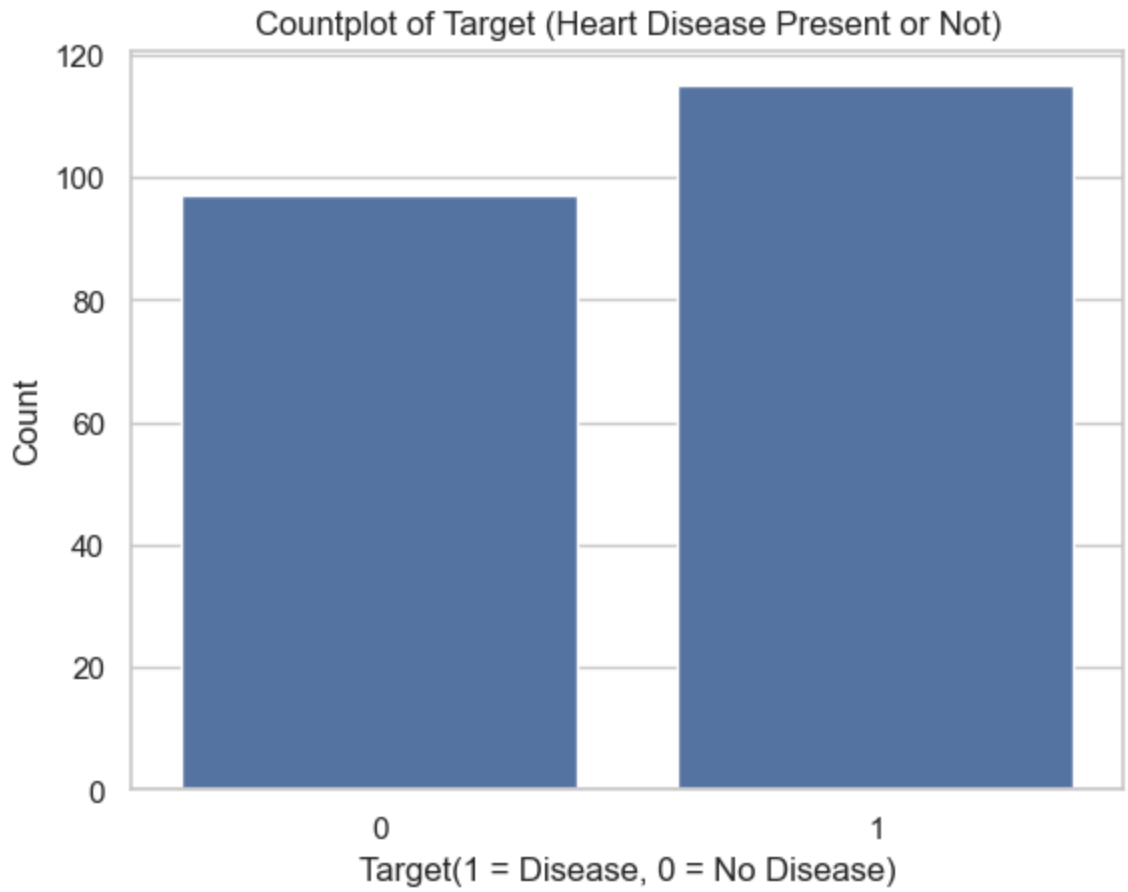
In [16]:
```
sns.boxplot(x='target', y='chol', data=heart)
plt.title("Boxplot of cholesterol by heart disease presence")
plt.show()
```



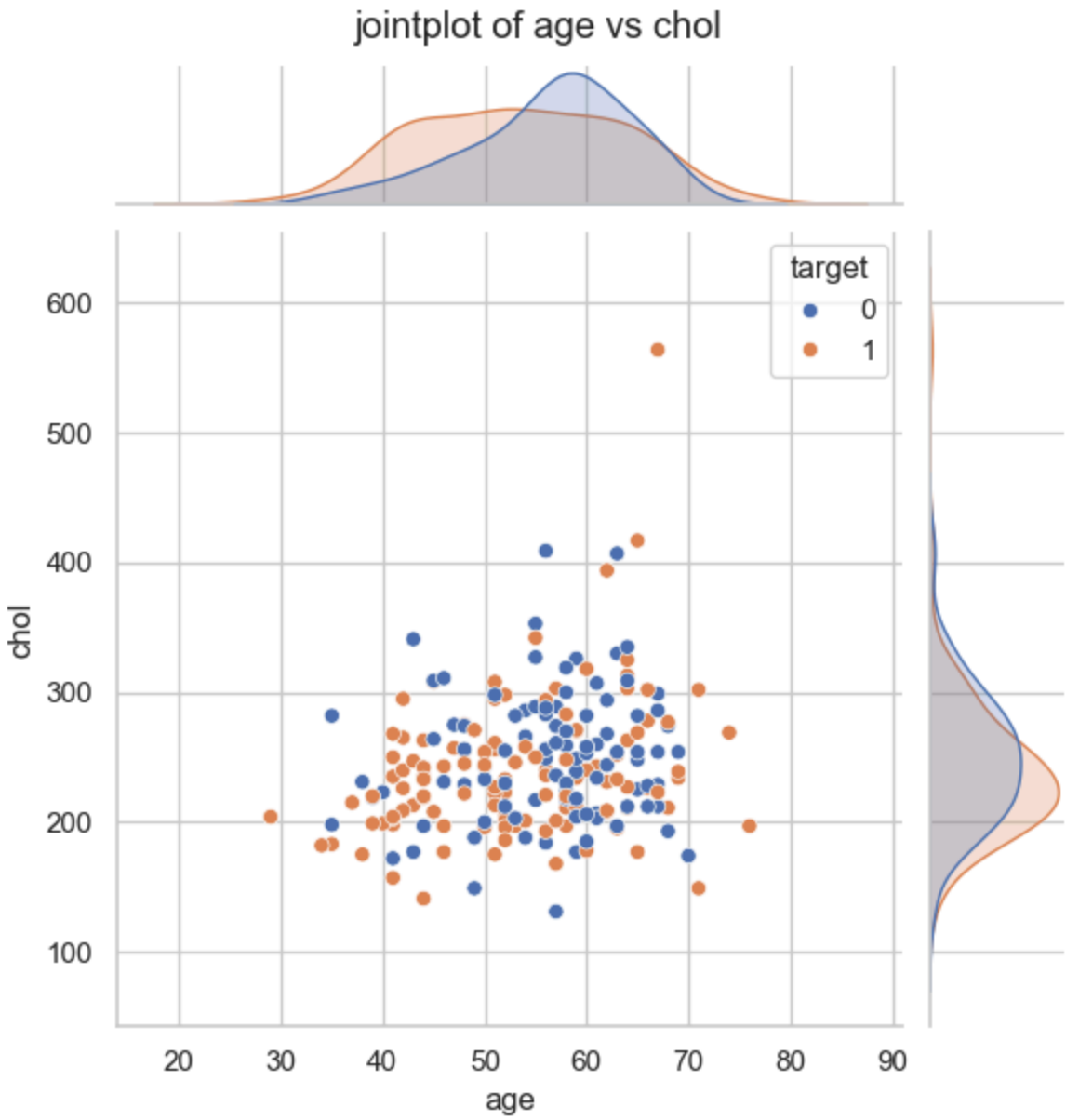
In [22]:
```
sns.pairplot(heart, hue='target')
plt.suptitle("Pairplot of Heart Dataset", y=1.02)
plt.show()
```

In [15]:

```
sns.countplot(x='binary_target', data=heart)
plt.title("Countplot of Target (Heart Disease Present or Not)")
plt.xlabel("Target(1 = Disease, 0 = No Disease)")
plt.ylabel("Count")
plt.show()
```

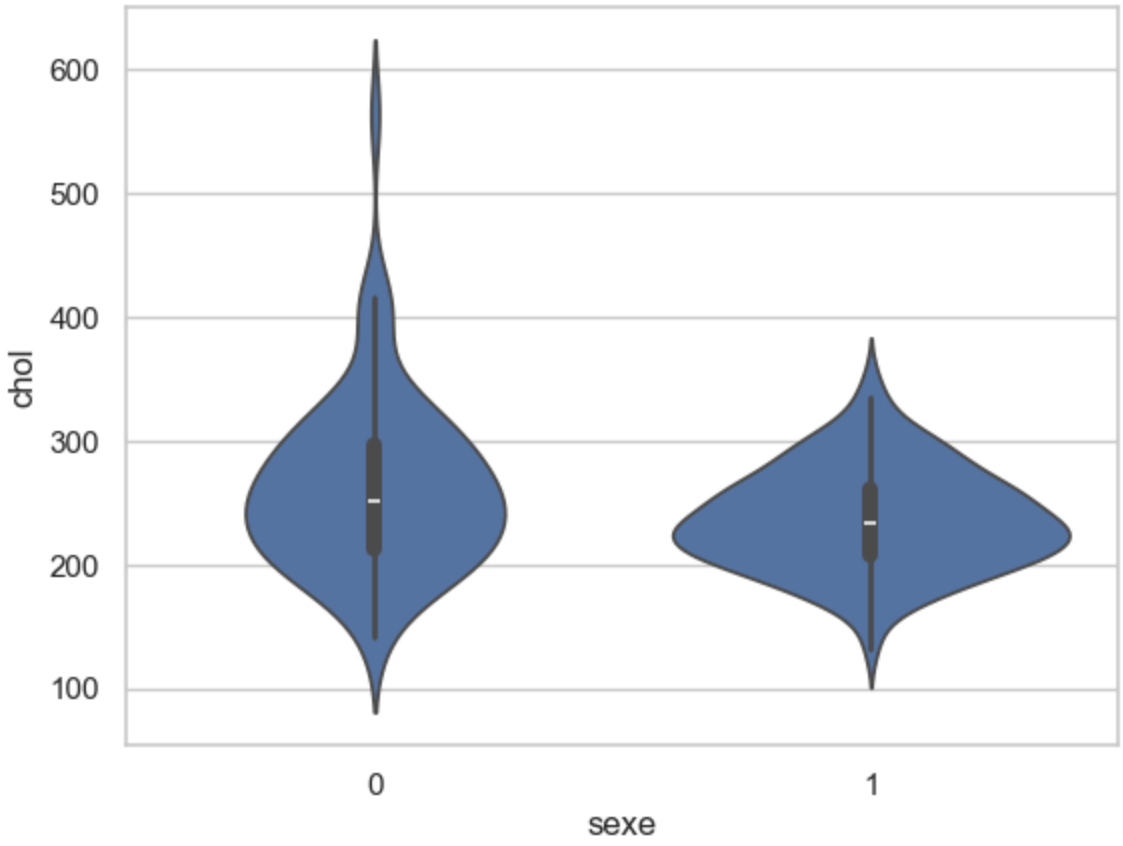


In [21]:
```
sns.jointplot(data=heart, x='age', y='chol', kind='scatter', hue='target')
plt.suptitle('jointplot of age vs chol',y=1.02)
plt.show()
```



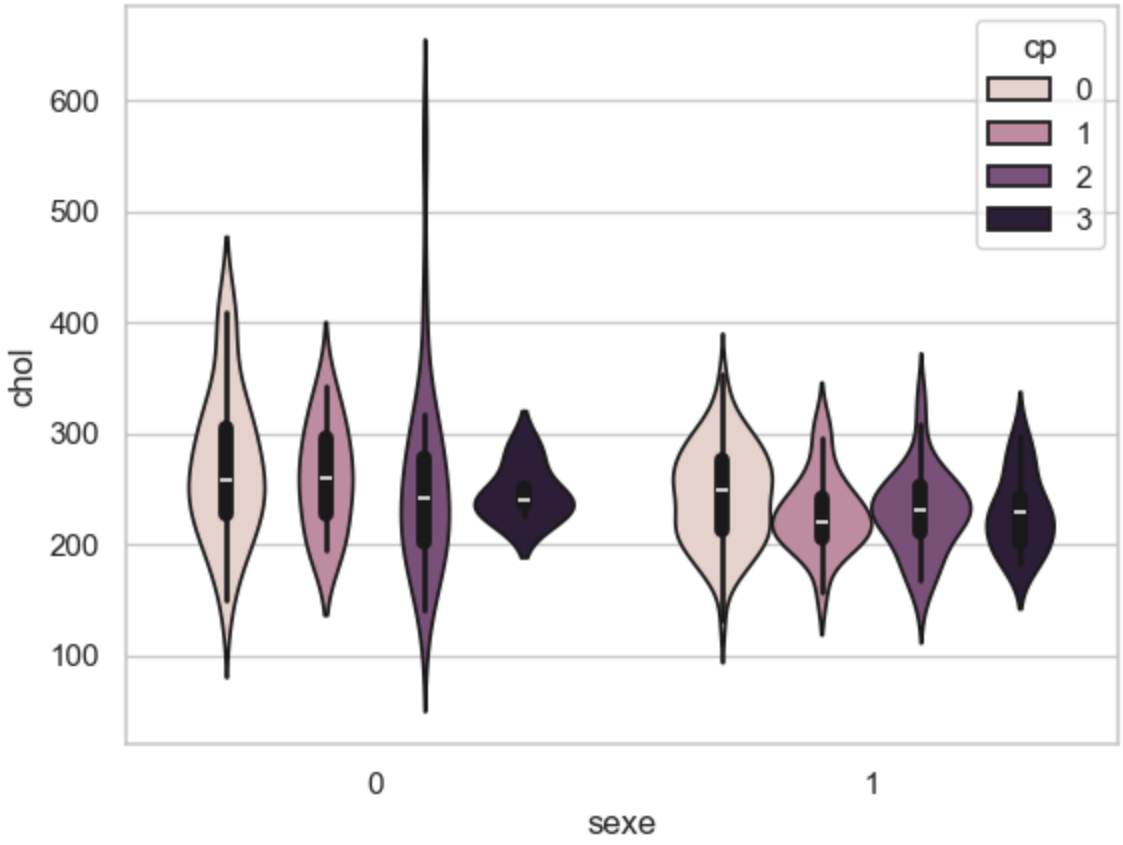
In [34]:
```
sns.violinplot(x='sexe', y='chol', data=heart)
```

Out[34]:
```
<Axes: xlabel='sexe', ylabel='chol'>
```



In [36]:
```
sns.violinplot(x='sexe', y='chol', data=heart, hue='cp')
```

Out[36]: <Axes: xlabel='sexe', ylabel='chol'>



In [ ]: